

## ORDER NOTATION QUESTIONS

28/02/26

1. A quadratic algorithm takes 2 ms to process 100 items of data.  
Roughly how long will it take to process 5000 items of data?
2. An algorithm with running time  $O(n \log n)$  takes 2 ms to process 100 items of data.  
Roughly how long will it take to process 5000 items of data?
3. Each of the expressions below represents the time complexity of a certain algorithm. For each, identify the dominant term, and hence give the lowest possible Big O complexity of the algorithm.

| Expression                            | Dominant term | $O(\dots)$ |
|---------------------------------------|---------------|------------|
| $5 + 0.001n^3 + 0.025n$               |               |            |
| $500n + 100n^{1.5} + 50n \log_{10} n$ |               |            |
| $n \log_3 n + n \log_2 n$             |               |            |
| $n^2 \log_2 n + n(\log_2 n)^2$        |               |            |
| $n^4 + 2^n$                           |               |            |
| $3n^3 + n!$                           |               |            |
| $n! + 3^n$                            |               |            |
| $(n + 1)! + n!$                       |               |            |

4. The statements below show some properties of Big O notation. Determine whether each statement is true or false, and correct the formula in the latter case.

| Statement                                                          | TRUE or FALSE? | Correction |
|--------------------------------------------------------------------|----------------|------------|
| Rule of sums:<br>$O(f + g) = O(f) + O(g)$                          |                |            |
| Rule of products:<br>$O(f \times g) = O(f) \times O(g)$            |                |            |
| Constants:<br>$O(kf) = O(f)$                                       |                |            |
| Transitivity:<br>If $g = O(f)$ and $h = O(f)$<br>then $g = O(h)$ . |                |            |

5. Algorithm X and Algorithm Y are different algorithms for solving the same problem.  
For a problem of size  $n$ ,
  - algorithm X has processing time  $f(n) = 0.1n^2 \log_{10} n$  microseconds
  - algorithm Y has processing time  $g(n) = 2.5n^2$  microseconds.
  - a. State the Big O time complexity of each algorithm.
  - b. Which algorithm will be better for very large problems?
  - c. Find a problem size  $n_0$  such that, if  $n \geq n_0$ , then the problem qualifies as “very large”.
  - d. If your problems are of size  $n \leq 10^9$ , which algorithm would be faster?
6. Algorithm P has complexity  $O(n \log n)$  and takes 10 microseconds to process 1024 items of data. Algorithm Q has complexity  $O(n^2)$ , but takes just 1 microsecond to process 1024 items of data. Which algorithm will be faster for processing  $2^{20}$  items of data?